

MATH 3670 Homework 1

Saahas Swaroop

September 4, 2025

1

Sample space with replacement:

$$S = \{rr, rg, rb, gg, gr, gb, bb, br, bg\}$$

Sample space without replacement:

$$S = \{rg, rb, gr, gb, br, bg\}$$

r : picking a red marble g : picking a green marble b : picking a blue marble
 rg : picking a red marble then a green marble

2

Sample space:

$$S = \{HHH, HHT, HTH, HTT, TTT, TTH, THT, THH\}$$

Event:

$$E = \{HHH, HHT, HTH, THH\}$$

3

a

$$\{7\}$$

b

$$\{1, 3, 4, 5, 7\}$$

c

$$\{3, 5, 7\}$$

d

$$\{1, 3, 4, 5\}$$

e

$\{4, 6\}$

f

$\{4, 7\}$

4

$$E = \{12, 14, 16, 21, 23, 25, 32, 34, 36, 41, 43, 45, 52, 54, 56, 61, 63, 65\}$$

$$F = \{11, 12, 13, 14, 15, 16\}$$

$$G = \{14, 23, 32, 41\}$$

$$EF = \{12, 14, 16\}$$

$$E \cup F = \{11, 12, 13, 14, 15, 16, 21, 23, 25, 32, 34, 36, 41, 43, 45, 52, 54, 56, 61, 63, 65\}$$

$$FG = \{14\}$$

$$EF^c = \{21, 23, 25, 32, 34, 36, 41, 43, 45, 52, 54, 56, 61, 63, 65\}$$

$$EG \cup FG = \{14\}$$

7

a

S

b

$\emptyset$

c

$\emptyset$

d

$\emptyset$

e

$$F \cup (E \cap G)$$

10

Let G be an event such that $F = E \cup G$ and $EG = \emptyset$. This means that the cardinality of F must be greater than or equal to that of E , as the smallest possible set G is $\emptyset$. Assuming all events have equal probability, $P(E) \leq P(F)$ due to their respective cardinalities.

14

The probability that exactly one of the events E or F occur is $P(E \cup F) - P(EF)$. Since $P(E \cup F) = P(E) + P(F) - P(EF)$, the original expression can be rewritten as $P(E) + P(F) - 2P(EF)$.

16

$\binom{n}{r} = \frac{n!}{r!(n-r)!}$ and $\binom{n}{n-r}$ can be rewritten as $\frac{n!}{(n-r)!(n-(n-r))!}$ which can be simplified to $\frac{n!}{n!(n-r)!}$, which shows both forms are equal.

An alternative method of proving this is by conceptually comparing what it means to choose r items from a group of n items. If we select r items, there will be $n - r$ items left behind. Instead of selecting r items in this way, we can instead select $n - r$ items, leaving r items behind. These are both equivalent, showcasing how $\binom{n}{n-r}$ and $\binom{n}{r}$ are equivalent.

18

a

$$\frac{1}{3}$$

b

$$\frac{1}{3}$$

c

$$\frac{1}{15}$$

23

$$P(R|R) = \frac{P(RR)}{P(R)} = \frac{\frac{1}{3}}{\frac{2}{3}} = \frac{1}{2}$$